AUTOMATION AND AUTONOMY

A designer must consider the autonomous capabilities needed to accomplish the mission of each robot built. The need to have a certain number of sensors, actuators, and algorithms arises from these capabilities.

- **Automation** refers to using technology to perform tasks or processes previously done manually by humans. It involves using machines, robots, or software to automate repetitive or labor-intensive tasks, increase efficiency, and reduce errors.
- **Autonomy** refers to the ability of a machine or system to perform tasks or make decisions without human intervention. Autonomous systems use sensors, artificial intelligence, and decision-making algorithms to operate and respond to environmental changes.

Four Sliders of Autonomous Capabilities

A. Plans: Generation versus Execution

Generating and executing a plan are closely related, as the robot cannot execute a plan unless one has been generated. The plan may be constructed by a human using a teach pendant to show the robot what the person wants it to do.

B. Actions: Deterministic versus Non-deterministic

Deterministic and non-deterministic are terms used in computer science to distinguish types of algorithms usually applied to finite state automata.

- **Deterministic FSA** – In each robot state and receives a specific set of inputs, there is only one possible output.
- **Non-deterministic** – involves multiple possible outputs, and the chosen output depends on some other factor or event.

C. Models: Open- versus Closed-World

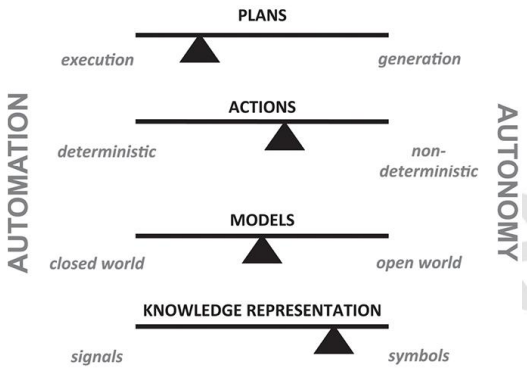
In robotics, a **world model** refers to the representation of the robot's environment in a computational form. It is a virtual model of the physical world that the robot operates in, which is created using sensors and other data sources.

- **Closed-world assumption** – assumes that everything outside of the known information is false or unknown. In other words, it assumes that the world is fully observable and known and that any information that is not available can be inferred or derived from what is known. A closed-world assumption is often used in formal logic and databases, where a query can be answered by checking the available information.
- **Open-world assumption** – assumes that the world is dynamic and that new information can always emerge. It assumes that the world is not fully observable and that unknown factors can affect the environment. In other words, an open-world assumption allows for the possibility that there may be information that is not yet available or unknown.

D. Knowledge Representation: **Symbols versus Signals** – In robotics knowledge representation, symbols and signals serve different roles in representing information.

- **Symbols**: Symbols are abstract representations that carry meaning. They often represent concepts, objects, or relationships between them.
- **Signals**: Signals, on the other hand, are physical phenomena that convey information. They are typically used to represent sensory input, such as the position of a robot's arm, the temperature of a room, or the sound of a voice.

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Impact on Types of Human Error

- **SUBSTITUTION MYTH** – The myth that a machine perfectly replaces a human for that task. "Autonomous" often implies to a user that the robot is more capable than it is, particularly in the subtle ways that make humans so adaptable.
- **HUMAN OUT-OF-THE-LOOP (OOTL) CONTROL PROBLEM** – From a larger human-machine perspective, no matter how independent of a human the robot is, if something goes wrong, a human is expected to take over for the robot or to intercede. However, it may not be cognitively possible for a human to react fast enough or to comprehend what the robot has been doing and what is going wrong and to determine why and then to correctly solve the problem.

Operational Architecture Attributes

- **Primitives** – Robots view autonomous capabilities such as SENSE, PLAN, ACT, and LEARN.
  - If a function takes in information from the robot's sensors and produces an output useful by other functions, then that function goes into the **SENSE** category.
  - If the function takes in information and produces one or more tasks for the robot to perform, that function is in the **PLAN** category.
  - Functions that produce output commands to motor actuators fall into **ACT**.
  - **LEARN** is an important term that transcends the other three primitives; an agent can learn to sense or plan better, acquire more acts or skills, or even learn for a specific mission what to sense, what to plan for and how to act.

Three robot primitives defined in terms of inputs and outputs.

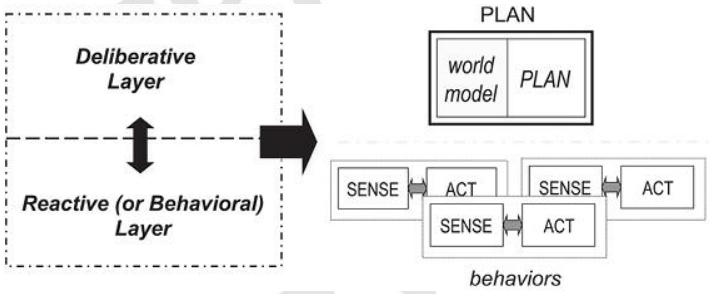
ROBOT PRIMITIVES	INPUT	OUTPUT
SENSE	Sensor data	Sensed information
PLAN	Information (sensed and/or cognitive)	Directives
ACT	Sensed information or directives	Actuator commands

- **Perceptual ability** – Robots view the perception needed for a particular capability as being either direct or requiring recognition.

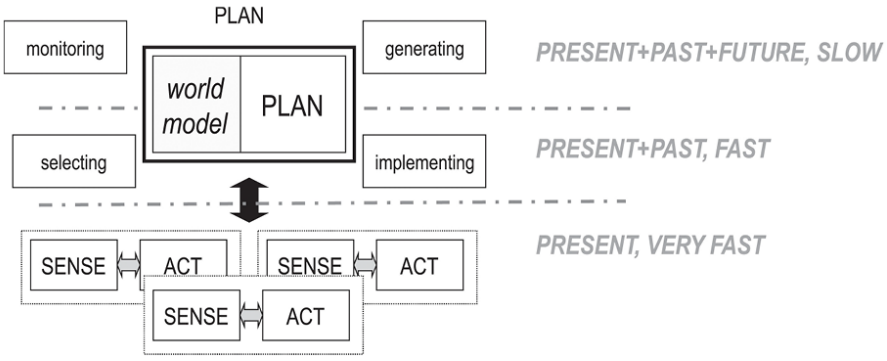
- **Planning horizon** – Robots view a capability as also having a planning horizon of either present, past/present, or future/past/present. It constrains the choice of data structures as well as algorithms.
- **Time-scale** – Robots consider the time-scale of a function: whether it needs to be very fast (a reflex), fast (selection of actions), or relatively slow (reasoning about a problem). It determines the action/operation of the software components.

Operational Architecture Layers

- **Reactive functionality (behavioral)** corresponds to the reactive loop in the central nervous system. This loop is associated with functionality in which it is based on responses and skills based on motor memory.
  - SENSE and ACT are tightly coupled into constructs called behaviors, which produce capabilities.



- **Deliberative functionality** corresponds to the cognitive loop associated with the cortex in the brain. The brain independently takes the input of the same signals used by the lower central nervous system and adds additional sensor processing to make conscious decisions.
  - GENERATING - generating PLANS which correspond to planning, reasoning, and problem-solving.
  - SELECTING - selecting specific resources to accomplish the PLAN, which corresponds to planning, resource allocation, and knowledge representation of capabilities.
  - IMPLEMENTING - implementing the PLAN, which corresponds to execution.
  - MONITORING - monitoring the execution of the PLAN to determine if it is meeting the goal, learning what is normal and anticipating potential failures.



- **Interactive functionality** is needed for interaction with other agents, either people, other robots, or software agents, which enables us to work together. The interaction may be regulated by

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reactive or cognitive loops. It may manifest themselves as changes in posture and gestures, proximity to each other, and verbal communication.

Levels of Operational Initiatives

- **No autonomy** – The robot follows rigid programming for performing a task or achieving a goal. A robot may always cover an opposing player in the same way or always pass the ball to another player.
- **Process autonomy** – The robot can choose the algorithm or process to achieve its task goals. The robot may adopt a different strategy for covering different players.
- **Systems-state autonomy** – The robot can generate and choose between options to achieve its goals. The robot might decide to keep the ball rather than pass the ball according to the playbook.
- **Intentional autonomy** – The robot can change its goals to meet the intent of its roles within the team, such as filling in for another player, where the intent to be a good team player is a higher priority than the intent to be a good midfielder. This functionality introduces the notion of being aware of others and explicitly interacting with them.
- **Constraint autonomy** – The robot can create its own roles and goals, which, in AI, may involve relaxing constraints on its goals or how it accomplishes its goals. At this highest level of initiative, a robot may ask itself if it even wants to play. Constraint autonomy reflects the level of initiative associated with robots in movies, where they suddenly decide to take over the world.

Telesystems

Telesystems enable humans and robots that are physically separated to interact to accomplish the task. It connotes that the human must interact with the world through the robot and would prefer to delegate tasks to the robot rather than micromanage every movement.

- A **taskable agent** domain is one where the robot is given a complex task or mission, executes it without supervision, and then returns or informs the human.
- A **remote presence** domain is one where the human and robot share the task, and the task execution is blended. This type of application is also called a **joint cognitive system** because the human and robot must jointly divide and share the cognitive activities to accomplish the mission.

Components of a Telesystem

- The **local** is where the human operator, or teleoperator, works.
  - A **display** is a component that presents information or data to the user. It can be a screen, monitor, or another visual output device that provides real-time feedback about the telesystem's status or performance. Displays can show various types of information, such as images, text, graphs, and video.
  - A **control mechanism** is a component that is used to operate and control the telesystem. It can be a button, switch, lever, or another input device that allows the user to interact with the telesystem. Control mechanisms are used to start, stop, adjust, or program the telesystem's behavior or actions.
- The **remote** (telefactor) must have **sensors**, **effectors**, **power**, and **mobility**.

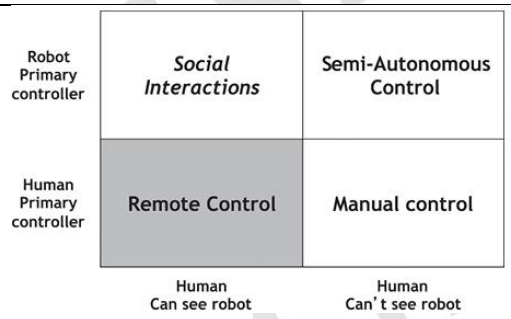
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- **Sensors** are devices or components that detect and measure physical properties or changes in the environment. They can detect various types of stimuli, such as temperature, light, sound, pressure, etc.
- **Effectors** are components that take action based on the information received from sensors or other sources. They act on the environment to produce a desired result or change. Examples of effectors include actuators, motors, and robotic arms.
- **Power** refers to the energy or electricity required to operate the telesystem. It is essential for the operation of the sensors, effectors, and other components of the telesystem.
- **Mobility** refers to the ability of the telesystem to move or travel in the environment. This can be accomplished through various means, such as wheels, legs, or propellers.

Types of Supervisory Control

Supervisory control is defined as when one or more human operators intermittently give directives and continually receive information from a computer that, itself, closes an autonomous control loop through artificial effectors and sensors to the controlled process or task environment.

- **Remote Control** - The operator can see the robot, and the robot has little, if any, intelligence.
- **Manual Control** - The operator cannot see the truly remote robot, and the robot has little intelligence.
- **Semi-Autonomy** - The robot has intelligence, but it may not have a set of capabilities that can totally substitute for human intelligence.
- **Social Interactions** - It covers situations where the human can see the robot and must interact with it, even though the human does not control the robot.



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